

TeamMatch

AI-Powered Team Formation & Progress Tracking

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Problem Statement

The Problem

- 1 Manual team formation is slow & biased
- 2 Groups end up skill-imbalanced
- 3 No visibility into weekly progress
- 4 Struggling students go undetected until it's too late
- 5 Instructors lack data to intervene early

Universities Affected

Most university project-based courses still rely on manual or semi-manual team assignment processes

Root Cause

Instructors must balance skills, schedules, experience, and communication styles without automated support, overall it's just outdated approach.

Consequence

Unbalanced teams can lead to uneven workloads, poor collaboration, and reduced accountability. Without structured progress tracking, instructors often cannot identify struggling students, communication issues, or missed milestones until late in the project lifecycle, when intervention becomes much harder.

Solution Overview



Smart Team Formation

Algorithm scores every possible team on skill balance, schedule overlap, and experience mix — assigns best combination



Weekly Check-In Tracking

Students submit hours, blockers, confidence level, and completion status each week for full accountability



Claude AI Health Reports

Anthropic Claude analyzes all check-in data and generates a structured 4-section health report per team



Analytics Dashboard

Instructors see line charts of team progress over time and can trigger AI analysis with one click



Notification System

Students receive alerts; instructors see at-risk flags surfaced from AI analysis of check-in patterns

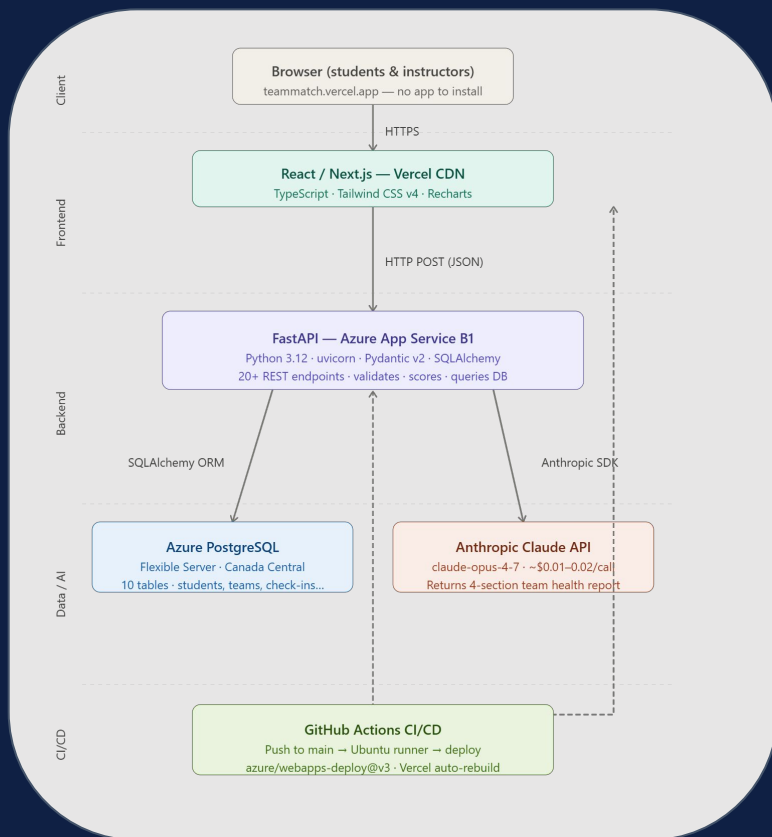


Full Cloud Deployment

Frontend on Vercel CDN, FastAPI backend on Azure App Service, PostgreSQL on Azure Flexible Server

System Architecture — Visual Overview

Real deployment stack · Azure App Service + Vercel CDN + Anthropic Claude API



Architecture Overview Full-stack platform connecting students and instructors through automated team matching and AI-powered insights.

System Components Browser → Next.js frontend → FastAPI backend → PostgreSQL & Claude API

Data Flow Representation All requests route through FastAPI — the frontend never touches the database or AI directly.

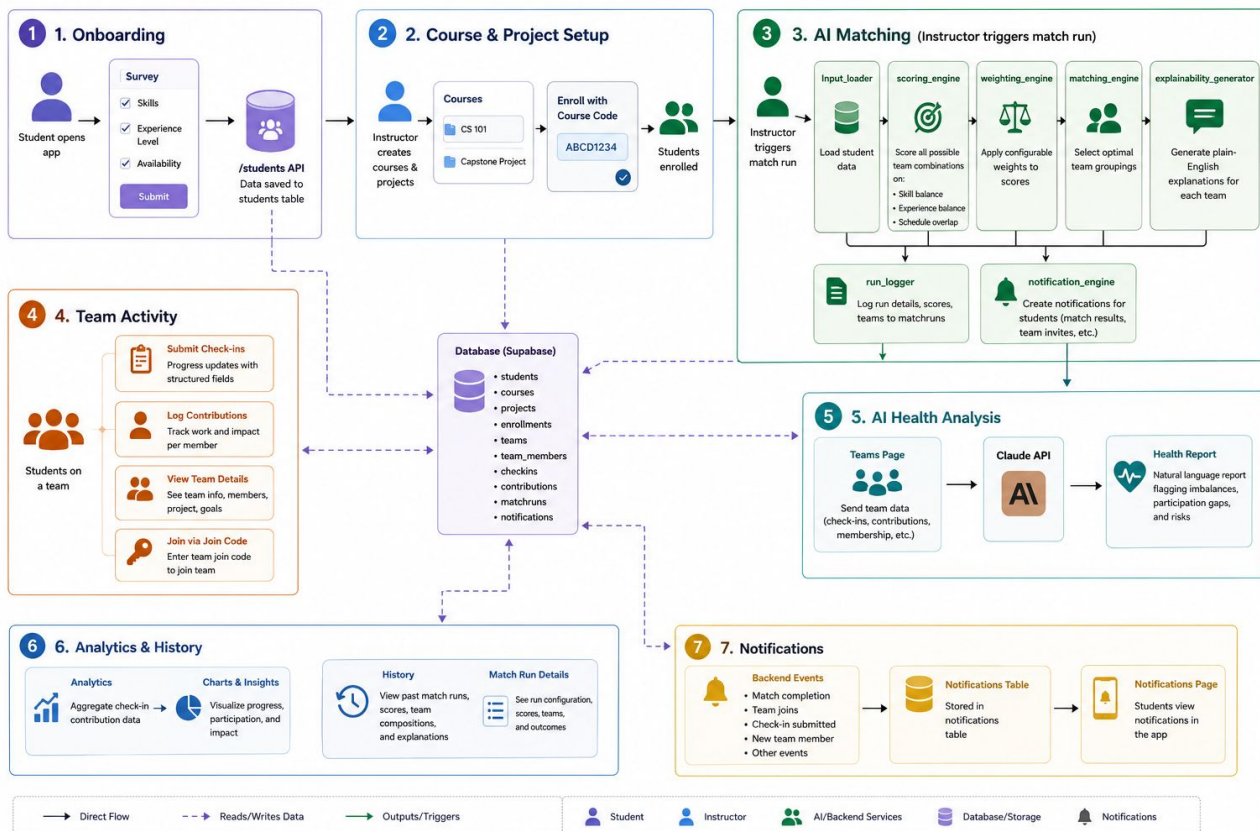
Technology Stack Legend React · Next.js · FastAPI · PostgreSQL · Anthropic Claude · Azure · Vercel

Database Design — Azure PostgreSQL

Table	Key Fields
courses	id, name, instructor_id, team_size, team_code
students	id, name, email, course_id, team_id, skills[], availability[], experience_level, leadership_preference
teams	id, name, team_code, course_id, skill_balance_score, schedule_overlap_score, experience_balance_score, overall_score
matchruns	id, course_id, status (PENDING/RUNNING/COMPLETED/FAILED), error_reason
checkins	id, student_id, team_id, week_number, hours_worked, completion_status, confidence_level, needs_help, blockers
projects	id, course_id, team_id, name, description, deadline, status
milestones	id, project_id, title, description, due_date, completed
contributions	id, student_id, team_id, description
notifications	id, student_id, message, read

All tables auto-created via SQLAlchemy Base.metadata.create_all() • Connected via Azure PostgreSQL Flexible Server

Database Design — Azure PostgreSQL



Design and Mockups

Application design for Teammatch V1

Instructor View:

TeamMatch API
AI-Powered Team Formation Tool

Login / Authentication

Email (required)

Remember me?

☐

Forgot password?

Login Button

[Don't have an account? Sign Up]

Screen Description

Login screen authenticates instructors. After successful login, user is directed to the Dashboard.

View Process Flow Diagram

TeamMatch API Dashboard

[Instructor Dr. Smith]

[View Flow]

[Logout]

3

Total Classes

137

Total Students

29

Teams Created

[Classes & Teams Overview]

[New Team Project]

[Class Name]	[Students]	[Teams]	[Status]	[Actions]
CS 101 - Intro to Programming	40	9	[Active]	[View Teams] [Edit]
CS 301 - Software Engineering	60	12	[Active]	[View Teams] [Edit]
CS 401 - Capstone Project	30	8	[Draft]	[View Teams] [Edit]

[Recent Activity]

Teams generated for CS 301 - 12 teams created

2 hours ago

Student data updated for CS 401 - 32 students

1 day ago

Teams reported for CS 101

3 days ago

[Contribution Tracking & Analytics]

[View Contribution Analytics]

[Team Health Dashboard]

[Student Join Teams]

[Student Check-in Status]

Screen Description

Dashboard shows overview of all classes and existing teams. Users can click "New Team Project" to start the team formation process (→ Upload Data screen or "View Teams" to see existing teams (→ Team Results screen).

Weekly Check-in

[Team Alpha - CS 301]

[View History]

[Weekly Check-in] - Track your contributions to help maintain team balance and accountability.

[Hours Worked This Week] *

[Tasks Completed] *

☒ Implemented login feature

[What I Worked On] *

Describe your contributions this week...

[Evidence Link] (Optional)

Link to GitHub commits, Drive documents, or other work evidence

[Blockers / Challenges]

Any obstacles or challenges you're facing...

[Peer Shoutout] (Optional)

Recognize a teammate's contribution...

[Cancel]

[Submit Check-in]

API Endpoint

POST /check-ins

Updates contribution score + team balance metrics

Screen Description

Weekly check-in form captures student contributions. Decision point "Data Valid?" validates required fields (hour, task, description). After submission, contribution scores update and instructor dashboard refreshes. If low contribution detected, system notifies student and flags instructor.

Legal, Security & Compliance

Data privacy · Licensing · Authentication · Vulnerability mitigation

Privacy & Ethics

- Collect only necessary data: skills, availability, course info
- No protected attributes (race, gender, religion) in matching
- Instructors can override or review any AI-generated team
- Users can edit or delete their data at any time
- Input validation prevents score manipulation
- Log suspicious behavior for admin review

Legal Compliance

- FERPA — academic data stays within institution
- GDPR / CCPA — PII minimized, deletion on request
- Full open-source stack: MIT / Apache 2.0 licenses
- React, FastAPI, SQLAlchemy, Next.js — all compliant
- AI team outputs owned by instructor / institution
- No copyrighted datasets used in matching logic

Security

- HTTPS / TLS for all data in transit
- JWT tokens with expiration + refresh
- Role-based access: student vs instructor
- Bcrypt password hashing
- All secrets in Azure App Settings — never hardcoded
- SQLAlchemy ORM prevents SQL injection; CSP blocks XSS

API Overview — 20+ Endpoints

Courses & Students

POST /students/

GET /students/course/{id}

GET /courses/{id}

AI & Analytics

POST /teams/{id}/analyze

GET /analytics/course/{id}

Team Matching

POST /matchruns/

GET /matchruns/{id}

GET /teams/course/{id}

Projects

POST /projects/

POST /projects/{id}/milestones

GET /projects/team/{id}

Check-Ins

POST /checkins/

GET /checkins/team/{id}

API #1 — Weekly Check-In | POST /checkins/

Purpose & Fields

- Students submit weekly progress reports
- Tracks hours worked, blockers, completion status
- Confidence level (1–5) flags struggling students
- Data feeds Claude AI team health analysis
- Instructor sees full team check-in history

Field	Type	Description
student_id	int	Required — links to student record
team_id	int	Required — links to team
week_number	int	1–15, current course week
hours_worked	int	Hours spent this week
completion_status	string	on_track / behind / blocked / ahead
confidence_level	int	1 (low) to 5 (high)
needs_help	bool	Flag for instructor alert
blockers	string	Free text — fed to Claude AI

POST /checkins/ — Request Body

```
{
  "student_id": 42,
  "team_id": 7,
  "week_number": 11,
  "hours_worked": 12,
  "completion_status": "on_track",
  "confidence_level": 4,
  "needs_help": false,
  "blockers": "None this week"
}
```

Response — 201 Created

```
{
  "id": 389,
  "student_id": 42,
  "team_id": 7,
  "week_number": 11,
  "created_at": "2026-05-05T19:22:00Z"
}
```

API #2 — Team Formation | POST /matchruns/

Algorithm Flow

1

POST /matchruns/ called by instructor

MatchRun record created — status: PENDING

2

Background thread starts run_matching()

Status transitions: PENDING → RUNNING

3

Score all team combinations

Skill balance + Schedule overlap + Experience mix

4

Select highest overall_score grouping

Weighted combination → 0.0 – 1.0 score

5

Assign students to teams

Status → COMPLETED; teams saved to DB

Request — POST /matchruns/

```
{  
  "course_id": 1  
}
```

Response — 201 Created (initial)

```
{  
  "id": 7,  
  "course_id": 1,  
  "status": "PENDING"  
}
```

GET /matchruns/{id} — Completed

```
{ "id": 7, "status": "COMPLETED",  
  
  "teams": [{ "name": "Team Alpha", "overall_score": 0.87 } ]  
}
```

JSON Request / Response — POST /teams/{id}/analyze

Request — Input Payload sent to Claude API

```
{
  "team_name": "Team Alpha",
  "overall_score": 0.87,
  "members": [
    {
      "name": "Alice Wang",
      "experience": "intermediate",
      "skills": ["Python", "React"],
      "checkins": [
        { "week": 9, "hours": 10,
          "status":
"on_track",
          "confidence": 4, "needs_help": false },
        { "week": 10, "hours": 6,
          "status":
"behind",
          "confidence": 2, "needs_help": true }
      ]
    }
  ]
}
```

Response — Claude AI Health Report (4 sections)

```
{
  "summary": "Team Alpha shows mixed progress.
  Alice dropped from week 9 to 10.",
  "at_risk_students": [
    {
      "name": "Alice Wang",
      "reason": "Confidence dropped to 2,
      behind on week 10, needs_help=true"
    }
  ],
  "recommendations": [
    "Schedule 1:1 check-in with Alice",
    "Review blocker from week 10"
  ],
  "action_plan": [
    "Week 12: Team sync on blockers",
    "Week 13: Milestone review"
  ]
}
```

Cloud API — Azure + Anthropic

Azure App Service

- FastAPI backend — B1 Basic, Python 3.12, Linux
- Region: Canada Central (same as DB = low latency)
- Auto-deploy: GitHub Actions CI/CD on every push to main
- Start cmd: `uvicorn app.main:app --host 0.0.0.0 --port 8000`
- `DATABASE_URL` & `ANTHROPIC_API_KEY` in Azure App Settings (env vars)

Azure PostgreSQL Flexible Server

- Server: `teammatch-server.postgres.database.azure.com`
- Tier: B1ms, region: Canada Central
- Schema auto-created by SQLAlchemy on startup
- Connected via `DATABASE_URL` environment variable
- Accessed only by backend — never directly exposed



Anthropic Claude API

- Called from FastAPI via Anthropic Python SDK
- API key stored in `ANTHROPIC_API_KEY` env var
- Model: `claude-opus-4-7` • ~\$0.01–0.02/call
- External cloud AI — not self-hosted



CI/CD Pipeline

- Push to main → GitHub Actions triggers
- Sets up Python 3.12, installs `requirements.txt`
- Deploys to Azure via `azure/webapps-deploy@v3`
- Vercel auto-rebuilds Next.js frontend in parallel

Testing & Improvements

Tested Feature	Issue Found	Fix Applied	Status
Match Run API	Runs stuck in PENDING after server restart	Added <code>_recover_stuck_runs()</code> on startup	✓ Fixed
Claude AI Analyze	JSON parse error on malformed AI output	Added try/catch + structured prompt	✓ Fixed
Check-In Submission	Missing field validation allowed nulls	Added Pydantic required field validators	✓ Fixed
Frontend Poll Status	COMPLETED state missed on fast runs	Increased polling interval + timeout fallback	✓ Fixed
Large Course Matching	35 students caused slow $O(n^2)$ matching	Added DB indexes on <code>course_id</code> , <code>team_id</code>	⚠ Partial
DB Connection Pool	Pool exhausted under concurrent load	Increased <code>pool_size</code> in SQLAlchemy engine	✓ Fixed

Peer testing conducted across all 3 demo courses with 100+ students — checks, match runs, and AI reports all verified end-to-end.

Metrics & Achievements

Projected outcomes · 3 demo courses · 100+ test students · full end-to-end validation

100+

Students Tested
across 3 demo courses

20+

REST API Endpoints
fully implemented

6 / 6

Critical Bugs Resolved
100% fix rate

\$0.01

Avg (mean) Claude AI Cost
per health report call

10

Database Tables
auto-schema on startup

4

System Layers
UI → API → DB → AI

Individual Contributions

Team APIs (all contributed): `/checkins/` • `/students/` • `/matchruns/` • `/teams/` • `/analytics/course/{id}`

Justin

Backend & Deployment

- Led FastAPI backend routes, models, schemas
- Azure App Service + PostgreSQL deployment
- GitHub Actions CI/CD pipeline
- **Individual:** `/matchruns/`, `/teams/{id}/analyze`, `/projects/`
- Environment config & Azure App Settings

Talin

Frontend & UX

- Built student-facing pages (survey, check-in, history)
- Instructor dashboard & teams views
- Tailwind CSS v4 styling & responsive layout
- **Individual:** `/notifications/`, `/history/` flows
- Vercel deployment & env config

Sako

AI & Analytics

- Anthropic Claude API integration
- Prompt engineering 4-section health report
- Analytics page with Recharts line charts
- **Individual:** `/teams/{id}/analyze`, `/analytics/`
- At-risk student flag logic
- Unit Testing

Zakir

Database & Schema

- Designed PostgreSQL schema (10 tables)
- SQLAlchemy ORM models & relationships
- Seeded demo data 3 courses, 100+ students
- **Individual:** `/students/`, **DB indexing & pooling**
- Startup recovery for stuck match runs

Viet

Testing & Integration

- End-to-end peer testing across all features
- Bug reports: validation, polling, JSON parsing
- Projects & milestones system
- **Individual:** `/projects/`, `/milestones/` endpoints
- Integration testing & check-in validation fixes

Future Improvements



Smarter AI Matching

Use Claude to suggest team formations based on past project outcomes, not just skill tags



Mobile App (React Native)

Students submit check-ins and view team progress from iOS/Android



GitHub Integration

Auto-pull commit history per student to augment check-in data with real contribution evidence



Predictive At-Risk Alerts

Train a lightweight model on check-in patterns to flag teams before they fall behind



Real-Time Messaging

In-app team chat channel with instructor oversight, replacing ad-hoc Slack/Discord



Multi-Instructor / LMS Integration

Canvas/Moodle webhook integration to sync course rosters automatically

Live Demo

TeamMatch is deployed — let's walk through it

- 1 Student survey → skill form → course enrollment
- 2 Instructor triggers match run → teams formed + scored
- 3 Student submits weekly check-in
- 4 Instructor clicks Analyze → Claude AI health report
- 5 Analytics dashboard → line charts per team

Thank you!